

Exercise 10: Constrained Optimization

*Lecturer: Aurelien Lucchi***Problem 1 (Constrained problem):**

Consider the following 2-dimensional problem

$$\begin{aligned} \min f(x, y) &:= x(1 - y^2) \\ \text{s.t. } x^2 + y^2 &= 1. \end{aligned}$$

1. Write the stationary and primal feasibility conditions.
2. Derive the optimal solution (x^*, y^*) .

Problem 2 (KKT problem with two constraints):

Consider the following 3-dimensional problem

$$\begin{aligned} \min f(x, y, z) &:= x + y + z \\ \text{s.t. } x^2 - y^2 &= 1 \text{ and } 2x + z - 1 = 0. \end{aligned}$$

1. Write the stationary and primal feasibility conditions.
2. Derive all the optimal solutions.
3. Can you comment on the results?

Problem 3 (Projection onto hyperplane):Consider the projection of a vector onto a hyperplane identified by the equation $\mathbf{Ax} = \mathbf{b}$, i.e.

$$\mathbf{x} = \text{Proj}_{\mathbf{Ax}=\mathbf{b}}(\mathbf{y}) = \underset{\mathbf{x}:\mathbf{Ax}=\mathbf{b}}{\text{argmin}} \frac{1}{2} \|\mathbf{y} - \mathbf{x}\|^2. \quad (1)$$

1. Write down the Lagrangian corresponding to the constrained problem defined in Eq. (1).
2. Calculate the optimal value of $\mathbf{x}$ (using the KKT conditions). Show that

$$\mathbf{x} = \mathbf{Py} + \mathbf{A}^\top (\mathbf{AA}^\top)^{-1} \mathbf{b},$$

where $\mathbf{P} := (\mathbf{I} - \mathbf{A}^\top (\mathbf{AA}^\top)^{-1} \mathbf{A})$ is a projection matrix.**Problem 4 (Normal cones):**

Consider the following two sets:

$$\Omega_\infty := \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_\infty \leq 1\},$$

and

$$\Omega_2 := \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 \leq 1\}.$$

1. Show that Ω_∞ and Ω_2 are non-empty, convex and closed.
2. Determine the normal cones of Ω_∞ and Ω_2 for $d = 2$ at the point $\mathbf{x} = (1, 0)$.